

# Oluwatobi Melvyn Mayungbo

Data Scientist, Machine Learning and Statistics

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## SUMMARY

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Data scientist focused on data quality and evaluation for AI. Built labelled evaluation sets for an LLM retrieval assistant and a crypto risk tool, a structure-aware document chunking method for LLM retrieval, and fraud models with careful handling of class imbalance. Works in Python with pandas, scikit-learn and Hugging Face.

## SKILLS

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**AI and LLM Systems:** LLM Evaluation and Guardrails, Hugging Face, Retrieval-Augmented Generation (RAG), Prompt Engineering, Citation Verification, Hybrid Search (BM25, Dense Embeddings, RRF), Model Context Protocol (MCP), ONNX Runtime

**Data Analysis:** Pandas, NumPy, SQL, Anomaly Detection

**Machine Learning:** Scikit-learn, PyTorch, LightGBM, XGBoost, Deep Learning, Sequence Models (GRU, TCN, Transformers), Model Calibration (Isotonic Regression, Temperature Scaling), SHAP, Feature Engineering

**Payments and Risk:** Web3 Fraud Risk Scoring, Transaction Fraud Detection, Credit Risk Decisioning, Payment Reconciliation, Double-Entry Ledgers and Idempotency, Regulatory Compliance (CBN/NDPA), Demand Forecasting

**Engineering and MLOps:** Python, SQL, DuckDB, PostgreSQL, Redis, FastAPI, Docker, GCP Cloud Run, AWS ECS Fargate, Railway, Prefect, dbt-core, MLflow, Evidently, CI/CD (GitHub Actions), Git, Pytest, Streamlit, React

**Statistics and Causal Inference:** Statistical Inference, Experiment Design, Power Analysis, CUPED, Causal Inference (S/T/X-Learners, Causal Forests), Uplift Modeling, Qini and AUUC, Time-Series Backtesting, Logistic Regression, WOE Scorecards

## PROJECTS

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### Fintech Compliance RAG Assistant | [GitHub](#) | [Live App](#)

Tech: Python, Streamlit, BM25, ONNX Runtime, Reciprocal Rank Fusion, FastAPI, Docker, GCP Cloud Run

- Engineered section-aware chunking that never merges across a document's own structural boundaries, keeping every citation attributable to exactly one clause, and validated it against a structure-blind fixed-window baseline rather than asserting it was better.
- Ran retrieval and generation evaluation as separate metrics to localise failure by component; traced a 0.40 refusal-correctness result to a lexical-overlap limitation in the offline stub judge and flagged re-validation against a live provider as the next step instead of overstating readiness.
- Built and deployed a regulatory question-answering system over central bank (CBN) circulars and the NDPA, combining sparse (BM25) and dense (ONNX-served MiniLM) retrieval fused via reciprocal rank fusion, reaching 0.96 recall@5 and 0.842 MRR on a 50-question labelled golden set with zero retrieval misses in the top 10.

### web3-risk-mcp: Web3 Fraud Risk MCP Server | [GitHub](#)

Tech: Python, Model Context Protocol (MCP), Etherscan V2, GoPlus, DexScreener, Docker, GitHub Actions

- Assembled an evaluation set of 34 hand-checked addresses (12 risky, 22 safe) with a record and replay harness reporting ROC AUC, precision and recall.
- Built an MCP server that gives AI assistants (Claude Desktop, Cursor) six read-only tools to check a crypto wallet, token or smart contract for risk before anyone interacts with it, returning a 0 to 100 risk score with a reason for every point, across Ethereum, Base, Arbitrum One, Polygon PoS and BNB Chain.
- Detects honeypot tokens, rug pull signs, hidden owner powers (mint, blacklist, pause), buy and sell tax, holder concentration, unlocked liquidity, and fund links to mixers, sanctioned wallets, exploiters and phishing addresses over 1 or 2 hops.

### Sequence-Based Transaction Fraud Detection System | [GitHub](#)

Tech: Python, PyTorch, LightGBM, ONNX Runtime, FastAPI, Docker, SHAP

- Handled extreme class imbalance with focal loss and weighted sampling and isolated their failure modes: focal loss degraded probability calibration (ECE 0.0063) while weighted sampling cost 0.042 PR-AUC. Corrected calibration with temperature scaling and isotonic regression, reducing ECE to 0.0005.

### Reckon: Payment Reconciliation and Review System | [GitHub](#) | [Live App](#)

Tech: Python, PyTorch, DuckDB, FastAPI, React, Docker, Railway, PostgreSQL

- Designed a human-in-the-loop review interface in React showing match confidence scores, audit trails and suggested resolution actions to streamline exception handling.

## EXPERIENCE

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### Machine Learning Instructor, SQI College of ICT

March 2026 to Present

- Teach machine learning and deep learning to student cohorts, covering statistical foundations, classical ML algorithms, neural architectures, sequence models and production evaluation metrics.
- Instruct students on data analysis, feature engineering and model building with PyTorch, Scikit-Learn, Pandas and NumPy.

### Internal Auditor, NISER

May 2024 to July 2024

- Analysed financial and operational records to identify inconsistencies and irregularities, performing manual anomaly detection across complex transactional datasets.

## EDUCATION

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### B.Sc. Statistics, University of Ibadan

January 2021 to March 2025

### Professional Diploma in Artificial Intelligence, SQI College of ICT

July 2025 to Present

## CERTIFICATIONS

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- Google Data Analytics Professional Certificate
- Google Advanced Data Analytics Professional Certificate
- Claude Code In Action